

The Relationship Between News Consumption Habits and Attitudes Toward Racial Groups

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Abstract

This report describes work done during an internship at CEU and the Complexity Science Hub in Vienna. The project had two parts: building a pipeline to turn free-text survey answers about news sources into usable variables, and running regressions to see whether those variables relate to attitudes toward different racial groups. The survey had 928 respondents who named their news sources across ten outlet types. We processed 2,771 non-empty cells through six cleaning steps. Splitting produced 6,372 candidate strings; sentence extraction added 1,079 rows; noise removal brought the total down to 6,885 clean rows covering 1,608 unique source names. Sources were assigned to one of eight content categories by an LLM and validated against human annotation (Cohen kappa = 0.736). Combining content categories with outlet groups gave 38 subcategory combinations, of which 18 had more than 50 respondents and were used in the regression. The regressions show that news consumption habits explain additional variance in racial attitudes beyond demographics. The biggest jump comes from adding the media predictors for attitudes toward Black people, where R^2 goes from 0.115 to 0.144. Political leaning and social circle are the strongest individual predictors overall.

AI Use Statement

Declaration of Generative AI

The following declaration outlines the usage of generative AI in the preparation of this internship report.

AI Tools Used

The following generative AI tools were used in the preparation of this report:

Claude (Claude Sonnet 4.5, Claude Sonnet 4.6): a large language model assistant developed by Anthropic, accessible via claude.ai.

Purpose of Usage

These tools were used for the following purposes:

Coding assistance: generating code, debugging, and refactoring scripts related to data processing, pipeline implementation, and visualization.

Writing assistance: proofreading and grammatical correction of report text, wording edits to improve clarity.

Source categorization: automated labeling of 1,680 unique news source names into eight content categories using a structured prompt. The categorization was validated against human annotation and achieved a Cohen kappa of 0.736.

Extent and Limitations

The use of generative AI was strictly limited to the purposes described above. I took full responsibility for all intellectual contributions central to this report, including the formulation of the research question, the design of the pipeline and regression analysis, the selection and review of methodology and literature, the structure of the report, and the interpretation and discussion of the results. All AI-generated outputs were reviewed, verified, edited, and corrected where necessary before being included in this report.

1. Introduction

1.1 General Background and Context

Racial attitudes—the beliefs, feelings, and behavioural predispositions that individuals hold towards members of different racial groups—are among the most extensively studied constructs in social psychology and political science. A long tradition of research has established that these attitudes are not formed in a vacuum: they are shaped by direct intergroup contact, by socialisation processes, and by the broader information environment in which individuals are embedded (Allport, 1954; Pettigrew & Tropp, 2006). As mass media has evolved from a handful of broadcast networks to an ecosystem of cable news, online outlets, social media platforms, podcasts, and independent digital creators, the routes through which information about racial groups reaches audiences have multiplied and diversified. How the structure of an individual's media consumption—which types of sources, through which platforms—relates to their attitudes towards racial outgroups is therefore both a substantive question of considerable societal importance and an empirical puzzle that has not been fully resolved in the existing literature.

This internship project was conducted at the Complexity Science Hub (CSH) and Central European University (CEU) in Vienna, Austria, under the supervision of Elisa Omodei and Clara Eminente. It formed part of a larger study examining racial misperceptions and their correlates in the United States (Eminente et al., 2026). The parent study collected a nationally representative survey of 866 U.S. respondents, measuring demographic characteristics, social circle composition, news and social media consumption habits, and attitudes towards four racial groups. Within this broader project, the present internship was tasked with two objectives: first, operationalising the respondents' free-text news source responses as structured predictor variables; and second, investigating the association between those variables and intergroup attitudes.

1.2 Statement of Purpose

The central purpose of this project was to analyse the relationship between news consumption habits—specifically the types of outlets and source categories through which respondents consume information—and attitudes towards different racial groups among White U.S. respondents. The raw survey data contained only free-text responses listing news sources, which could not directly serve as regression predictors. A substantial part of the project therefore involved constructing a reproducible text-processing pipeline to transform those responses into a structured matrix of outlet-by-source-category subcategory scores, suitable for statistical modelling. These scores were then used in a nested regression framework to test whether differences in media consumption patterns explain variance in racial attitudes beyond that attributable to sociodemographic characteristics.

1.3 Significance in a Broader Context

Misperceptions about the size of minority populations and negative attitudes towards racial outgroups have been linked to reduced support for equity-promoting policies, heightened perceptions of threat, and social division (Lawrence & Sides, 2014; Bursztyn & Yang, 2022). Understanding the media pathways that contribute to or attenuate such attitudes is therefore of direct policy relevance. Moreover, prior research has largely examined news consumption as a unidimensional variable (e.g., how often one watches television news), without distinguishing between the type of content consumed or the platform through which it is accessed. This project addresses that gap by constructing a multidimensional operationalisation of media consumption that captures both outlet type and source category, offering a more granular test of information environment effects on intergroup attitudes.

1.4 Report Structure

The remainder of this report is structured as follows. Section 2 reviews the relevant literature and derives the research question. Section 3 describes the data and methodology, including survey design, the text-processing pipeline, LLM categorisation, validation, and regression modelling. Section 4 presents results. Section 5 discusses findings in relation to the research question and

existing literature, acknowledges limitations, and makes recommendations for future work. Section 6 concludes.

2. Literature Review

2.1 Intergroup Attitudes and Their Determinants

The study of intergroup attitudes—how members of one social group perceive, evaluate, and feel toward members of another—has been a central preoccupation of social psychology since at least Allport's (1954) landmark formulation of the contact hypothesis. Allport argued that direct, cooperative, equal-status contact between members of different groups reduces prejudice by correcting factual misperceptions and promoting perspective-taking. A meta-analysis by Pettigrew and Tropp (2006) confirmed this prediction across 515 studies: intergroup contact reliably reduces prejudice, with a mean effect size of $r = -0.21$. However, direct contact is only one of several determinants of intergroup attitudes. Structural factors—including residential segregation, occupational inequality, and historical discrimination—shape the social contexts in which attitudes develop, while individual-level factors such as education, political orientation, and authoritarianism moderate how those contexts are processed (Sidanius & Pratto, 1999; Kinder & Sears, 1981).

A particularly robust finding in the U.S. context is the partisan structuring of racial attitudes. Since at least the Civil Rights era, racial conservatism and Republican identification have been closely linked, while Democratic identification has been associated with more positive attitudes towards racial minorities (Kinder & Sanders, 1996; Tesler, 2016). This partisan polarisation has intensified in recent decades, as racial and ethnic issues have become increasingly sorted along party lines (Sides, Tesler, & Vavreck, 2018). Any analysis of media effects on racial attitudes must therefore account for the strong confound introduced by political identity.

2.2 Media Exposure and Intergroup Attitudes

The role of media in shaping public perceptions of racial and ethnic minorities has been studied across multiple decades and methodological traditions. A consistent finding from content analyses is that mainstream U.S. news media over-represent racial minorities—particularly Black individuals—in crime-related coverage, while under-representing them in non-crime news (Dixon & Linz, 2000; Travis & Leechman, 2014). Experimental studies have shown that exposure to such crime-heavy portrayals increases viewers' perceptions of Black criminality and fear of crime (Dixon, 2008; Chiricos, Hogan, & Gertz, 1997). Conversely, exposure to positive or counter-stereotypical representations of minority group members has been found to reduce implicit and explicit biases (Bodenhausen, Schwarz, Bless, & Wanke, 1995; Schiappa, Gregg, & Hewes, 2005).

Beyond the content of coverage, the medium through which information is consumed matters. Scholars working in the tradition of cultivation theory (Gerbner & Gross, 1976; Gerbner et al., 2002) have argued that heavy television consumption cultivates distorted perceptions of social reality, including inflated estimates of crime and over-representation of minority groups. More recent work has extended this framework to online and social media environments, finding that individuals who rely primarily on social media for news hold less accurate demographic perceptions (Eminente et al., 2026; Ahmed & Rasul, 2022). Herda (2010), studying overestimation of immigrant population sizes across twenty-three European countries, found that those who consumed traditional news more frequently held more accurate perceptions, consistent with the idea that structured exposure to fact-checked information corrects misperceptions. In contrast, high social media use has been associated with higher rates of overestimation of minority group sizes, potentially because algorithmically curated feeds expose users to a more ethnically diverse—but statistically unrepresentative—sample of content (Eminente et al., 2026).

The political slant of news sources adds further complexity. Prior research has established that exposure to ideologically congruent news reinforces prior attitudes, while exposure to counter-attitudinal content can shift them, though effect sizes are typically small (Levendusky & Malhotra, 2016; Stroud, 2011). With respect to racial attitudes specifically, Althaus and Kim (2006) found

that political talk radio—predominantly conservative in the U.S. context—was associated with less positive attitudes towards affirmative action, while exposure to liberal print media was associated with more positive attitudes. This suggests that the ideological valence of consumed media is a relevant moderator of attitude formation, though the causal direction of the relationship remains difficult to establish in cross-sectional data.

2.3 Social Media, News Creators, and the Fragmented Information Landscape

The rise of social media platforms and independent digital creators (variously called “news creators,” “citizen journalists,” or “influencers”) has fundamentally altered the media landscape in which attitudes form. Unlike traditional broadcast media, social media platforms rely on algorithmic curation that optimises for engagement, which tends to promote emotionally salient and identity-confirming content (Pariser, 2011; Bail et al., 2018). Research has found that individuals who rely heavily on social media for news are more likely to encounter and share misinformation (Ahmed & Rasul, 2022; Ahmed & Rasul, 2023), and that social media use is associated with higher rates of overestimation of minority group sizes (Eminente et al., 2026). The mechanism proposed is that the algorithmically curated diversity of social media feeds creates an “illusion of diversity”—a perceived social environment more ethnically varied than the user’s actual community or the national population, which inflates estimates of minority group size and may affect attitudes through perceived group threat or salience.

At the same time, YouTube and other long-form platforms host a range of independent news creators whose content varies widely in ideological orientation, analytical depth, and factual accuracy. Unlike broadcast news organisations, these creators are not subject to editorial standards or regulatory oversight, and their content can range from careful analytical commentary to partisan advocacy or misinformation. The distinction between consuming a mainstream national news channel through television versus through YouTube is therefore analytically meaningful: the former involves scheduled, editorially selected programming, while the latter involves algorithmically recommended content from a mix of professional and amateur producers. This distinction motivates the outlet-by-category operationalisation developed in the present project.

2.4 Research Gap and Research Question

Despite this substantial literature, several gaps remain. First, most studies of media effects on racial attitudes have treated news consumption as a unidimensional variable, typically measuring only overall frequency or platform usage without capturing what type of content is consumed or from what category of source. Second, the distinction between platform of access and source type has rarely been operationalised systematically: whether one encounters The New York Times via a print subscription, a website, a social media share, or a podcast implies different consumption behaviours and social contexts, yet most studies collapse these distinctions. Third, the growing importance of social media-native news creators as a distinct information source type has received little attention in the racial attitudes literature.

The present project addresses these gaps by constructing a multidimensional characterisation of news consumption in which each respondent’s media diet is described along two dimensions simultaneously: the outlet group through which content is consumed (TV/Radio, Print/Online, Social Media, YouTube, Podcast) and the content category of the source (National News Channel, Print & Digital News Outlet, News Creator, Local News, Radio and Podcasts, News + Entertainment, Journalists & Hosts, Non-News Related). The resulting outlet-by-category subcategory variables capture the texture of respondents’ information environments in a way that prior measures have not. The research question guiding the analysis is:

What is the relationship between the type and outlet of news sources consumed and attitudes toward different racial groups, controlling for sociodemographic characteristics and direct social contact?

3. Methodology

3.1 Research Approach

The project comprised two major parts. The first part operationalised news consumption diversity by developing a text-processing pipeline to extract, clean, categorise, and structure the free-text news source data. We justified this approach on the grounds that free-text responses capture the authentic richness of respondents' media diets—including sources that would not appear on a pre-specified list—while requiring systematic processing to become analytically usable. The second part used the resulting structured variables as predictors in a nested OLS regression framework to test the relationship between news consumption patterns and intergroup attitudes. We selected OLS regression over more complex models because it provides interpretable coefficient estimates and standard errors, enabling assessment of both the magnitude and statistical significance of each predictor's contribution. The nested structure, in which sociodemographic predictors are entered first and media predictors added in a second step, allows us to quantify the incremental explanatory power of news consumption variables beyond the variance explained by demographic controls.

3.2 Data Source and Collection

The data originated from a purpose-built survey of U.S. participants administered between May and June 2024 through the online platform Prolific. The survey was designed to be nationally representative by gender, age, and race. Respondents were recruited for a target of 1,000 participants. Following data cleaning—which excluded respondents with serious internal inconsistencies in their answers (e.g., mismatches between reported zip code, city, and state of residence), those who identified as non-binary (approximately 2% of the sample), those who did not disclose their political orientation (approximately 3%) or income (approximately 1.7%), and those who selected “Other” as a political leaning option (5 respondents)—the final analytical sample consisted of $N = 866$ respondents. Of these, 928 provided usable responses to the news source questions and form the base of the media consumption analysis.

The survey first collected standard sociodemographic information: gender (Female/Male), age (retrieved from Prolific and binned into five categories: 18–24, 25–34, 35–44, 45–54, 55+), education (six levels from “some high school or less” to “graduate or professional degree”), annual income (six levels from “less than \$25,000” to “\$150,000 or more”), and employment status (collapsed to full-time versus not full-time). Respondents also reported their political leaning (Democrat, Republican, Independent). Before proceeding to the main survey tasks, respondents completed a numeracy training section in which they practised converting fractions to percentages; only respondents who successfully completed this training were retained, ensuring a minimum level of comfort with proportional reasoning.

Respondents then estimated the percentage of different racial groups (White, Hispanic, Black, Asian, and Other) in their social circle, neighbourhood, city, state, and the United States as a whole. After each estimate, they reported their confidence on a 1–6 scale. They also reported the percentage of their social circle belonging to each racial group, the perceived frequency of news coverage of people of color (expressed as a percentage of total coverage), and the perceived tone of that coverage on a 1–6 scale (higher values = more positive). Finally, respondents reported which news sources they use, organised across ten outlet columns: TV news programs, online news sites, podcasts, printed newspapers, the radio, Facebook, Instagram, TikTok, Twitter/X, and YouTube. All news source fields were free-text, allowing respondents to write any response. This is the data on which the present project focuses.

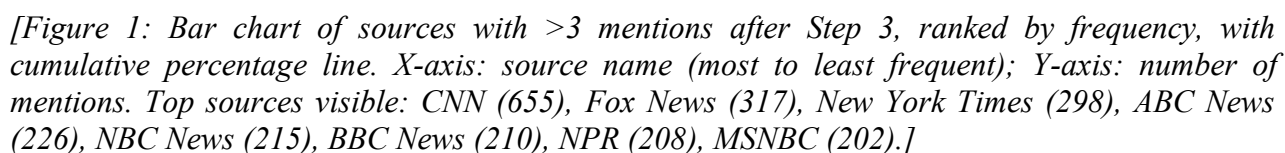
Racial attitudes were measured using five questions assessing respondents' attitudes towards each of four racial groups (Black, White, Asian, Hispanic). Three questions were race-specific (e.g., “How comfortable would you be if your child married someone of [race]?”) and two concerned racially integrated societies in general. Responses were on a 1–6 scale and averaged, then reversed so that higher values indicate greater tolerance. A reduced version—used as the outcome variable throughout—was computed as the average of the two race-specific questions about marriage and workplace interactions only.

We developed a six-step text-processing pipeline, implemented in Python using the pandas and re (regular expressions) libraries, to transform the free-text news source responses into structured data. The pipeline was designed to be reproducible and is fully documented in the accompanying Jupyter notebook. Each step is described below.

Each survey cell potentially contained multiple source names separated by commas, semicolons, newlines, slashes, or the word “and.” We developed a splitting function that decomposed each cell into individual candidate source strings. The conjunction “and” required special handling because it appears in legitimate source names (e.g., “Beau of the Fifth Column,” “Breaking Points with Krystal and Saagar”). We addressed this by (a) maintaining a protected-names list that was never split, and (b) splitting on “and” only when the left-side segment was 30 characters or fewer in length, making it likely to be a short source name rather than a full sentence. After splitting, 928 respondents across 10 outlet columns yielded a total of 12,051 non-empty cell entries.

Each candidate string was normalised by converting to lowercase, stripping URLs and domain suffixes (.com, .org, .co.uk, etc.), standardising punctuation (replacing underscores with spaces, removing @, trademark symbols, and parentheses), removing non-ASCII characters, and stripping leading and trailing whitespace and punctuation. This step ensured that “CNN,” “cnn,” “cnn.com,” and “www.cnn.com” all resolved to the single cleaned string “cnn,” enabling accurate frequency counting and alias matching in subsequent steps.

We constructed a frequency table counting how many times each unique cleaned string appeared across all respondents and outlet columns. This revealed a highly skewed distribution typical of Zipf-distributed named-entity data: a small number of source names (CNN: 655 mentions, Fox News: 317, New York Times: 298) account for a disproportionate share of total mentions, while the vast majority of strings appear only once or twice. Strings appearing more than three times were designated as “frequent sources” (161 unique strings) and became the focus of subsequent processing steps. These frequent sources collectively covered 57.1% of all total mentions. Figure 1 below shows the distribution of mention counts for these frequent sources, with a cumulative percentage overlay illustrating the concentration of mentions among the most commonly cited outlets.



Some respondents embedded source names within sentences rather than writing them directly (e.g., “I get news from CNN and BBC” or “entertainment news from figures like Arnold”). We

developed a name extraction function that searched each cleaned string for occurrences of any frequent source name as a whole word boundary match (using regular expressions), and replaced the full sentence with the extracted names. When multiple source names were found in a single string, we exploded the string into multiple rows, one per extracted source. This step recovered previously inaccessible source mentions that would otherwise have been lost as long junk strings. After extraction, the frequent sources covered 70.1% of all total mentions, an increase of 13 percentage points over Step 3, confirming that the extraction rescued a non-trivial quantity of mentions.

[Figure 2: Bar chart of sources with >3 mentions after Step 4, ranked by frequency, with cumulative percentage line. Layout identical to Figure 1, showing the increase in coverage of frequent sources.]

A comprehensive alias dictionary was applied to all sources, mapping spelling variants, abbreviations, typos, and platform-prefixed names to a single canonical form. The alias map contained over 400 pattern-to-canonical mappings and was constructed by systematically reviewing the frequency table from Step 4 and consolidating variants. For example, “fox,” “fox news,” “foxnews,” “brietbart,” and “breibart” were all resolved to their canonical forms. Platform prefixes such as “youtube – ” and “tiktok: ” were stripped before alias matching. Generic noise strings—including “n/a,” “various,” “my feed,” “no specific sources,” and similar non-informative responses—were flagged as NOISE and removed. After unification, the frequent sources covered 75.2% of all total mentions, and the total set of unique sources was reduced from over 2,000 pre-unification strings to 1,835 canonical names. The total number of clean mention rows was 6,213.

[Figure 3: Bar chart of sources with >3 mentions after Step 6 unification, ranked by frequency, with cumulative percentage line. Layout identical to Figures 1 and 2, illustrating further concentration of mentions under canonical names.]

We submitted the final list of 1,680 unique canonical source names to Claude Sonnet (Anthropic) for categorisation using a detailed prompt developed and refined over six iterations. The prompt instructed the model to assign each source to exactly one of eight categories:

1. National News Channel: broadcast and cable TV news channels and networks that produce and distribute national or international news programming.
2. Journalists and Hosts: named individuals employed by or closely associated with a specific news organisation.
3. Print and Digital News Outlet: newspapers, magazines, and digital-native news publications.
4. Local News: local television stations, local newspapers, and local radio stations identified by call letters or city name.
5. Radio and Podcasts: radio networks and podcasts not classified as Local News.
6. News and Entertainment: programs or personalities that blend news commentary with entertainment formats.
7. News Creator: independent digital creators, YouTubers, and social media personalities producing original news commentary.
8. Non-News Related: entertainment platforms, sports media, social media platforms, and any source not primarily concerned with news.

Each category was defined in detail with explicit positive examples and, in later prompt iterations, explicit negative examples and disambiguation rules to handle boundary cases. Key disambiguation rules included: the distinction between a named individual currently employed by a broadcast network (Journalists and Hosts) versus the same individual operating independently (News Creator); the classification of local radio call-letter stations as Local News rather than Radio and Podcasts; and the classification of news aggregator websites as Non-News Related. Prompt development followed an iterative validation approach: we ran the prompt on the full dataset, manually checked a sample of outputs, identified errors, and revised the prompt. Six iterations were completed before we accepted the final categorisation.

The resulting category distribution across 1,680 sources was: Non-News Related (612 sources, 36.4%), Local News (330, 19.6%), News Creator (315, 18.8%), Radio and Podcasts (205, 12.2%), Print and Digital News Outlet (120, 7.1%), National News Channel (44, 2.6%), News and Entertainment (31, 1.8%), and Journalists and Hosts (23, 1.4%).

3.5 Validation

To assess the quality of the LLM categorisation, we drew a stratified random sample of 50 sources, stratifying by LLM-assigned category proportional to the category distribution in the full dataset. Both human annotators (the author and supervisor) independently assigned a category to each of the 50 sources using the final prompt as a reference guide, without viewing the LLM’s assignments. We measured inter-rater agreement using raw percentage agreement and Cohen’s kappa (κ), which corrects for chance agreement.

Human annotators agreed on 34 of the 50 sources from the outset. For the 16 sources on which they initially disagreed, a consensus category was determined through discussion. Cohen’s κ between the human consensus and the LLM’s categories was 0.736, indicating substantial agreement and exceeding our pre-specified threshold of 0.6 for acceptable categorisation quality. Per-category agreement results are shown in Table 1.

LLM Category	N	Agreements	Raw %	Cohen's κ
Local News	10	10/10	100.0%	1.000 (Almost perfect)
Journalists & Hosts	1	1/1	100.0%	1.000 (Almost perfect)
Non-News Related	18	16/18	88.9%	0.669 (Substantial)
Radio and Podcasts	6	4/6	66.7%	0.694 (Substantial)

Print & Digital News Outlet	4	2/4	50.0%	0.648 (Substantial)
News Creator	9	6/9	66.7%	0.646 (Substantial)
National News Channel	1	1/1	100.0%	0.658 (Substantial)
News + Entertainment	1	0/1	0.0%	0.000 (Slight; n=1)

Table 1. Inter-annotator agreement between human consensus and LLM categorisation, stratified by LLM-assigned category.

Local News and Journalists & Hosts achieved perfect agreement, though the latter was based on a single sampled source. Non-News Related, News Creator, Radio and Podcasts, Print & Digital News Outlet, and National News Channel all showed substantial agreement (κ 0.646–0.694). The only weak result was for News + Entertainment (κ = 0.000), again based on a single source (Gutfeld!), where the human annotators assigned the program to National News Channel while the LLM assigned it to News + Entertainment—reflecting genuine definitional ambiguity around politically opinionated cable programs. Given that all per-category κ values were either substantial or based on a single source, we considered the overall categorisation reliable and retained it for downstream analysis.

3.6 Subcategory construction

3.6 Subcategory Construction

We constructed subcategory predictors by cross-tabulating each source’s LLM-assigned category with the outlet group through which it was consumed. We defined five outlet groups based on theoretical and empirical considerations:

- TV/Radio: TV news programs and the radio, grouped because both involve scheduled broadcast consumption from a curated, non-algorithmic programming stream.
- Print/Online: printed newspapers and online news sites, grouped because both primarily involve text-based article reading and because the same types of sources (e.g., New York Times) appear across both columns.
- YouTube: treated separately from other social media because it primarily hosts long-form video content and is the dominant platform for News Creator consumption, with algorithmically recommended viewing rather than feed-based scrolling.
- Social Media: Facebook, Instagram, TikTok, and Twitter/X, grouped as feed-based short-form platforms where content is encountered passively through algorithmic curation.
- Podcast: audio-on-demand content, distinct from radio because of its asynchronous, subscription-based consumption model.

This distinction between outlet groups matters analytically: a respondent who encounters CNN through television is exposed to an editorially scheduled broadcast, while a respondent who encounters CNN through social media encounters a clip that has been algorithmically selected and may be embedded in a very different informational context. The same source consumed via different outlets implies different media behaviours and different potential influence pathways.

Subcategories were formed from all combinations of the five outlet groups and eight content categories, yielding up to 40 possible subcategories. We excluded combinations where fewer than 50 unique respondents mentioned any source, to ensure sufficient power for regression analysis. This threshold retained 20 subcategories as predictors. For each respondent, we created two versions of the subcategory variables: (1) a count variable indicating how many unique sources of that subcategory the respondent mentioned, and (2) a binary variable (0/1) indicating whether the respondent mentioned at least one source of that subcategory.

Figure 4: Unique respondents per subcategory (threshold = 50 shown in red)

Subcategory	Unique Respondents (approx.)
TV/Radio National News Channel	430
Print/Online Print & Digital News Outlet	340
Print/Online National News Channel	315
Social Media Non-News Related	255
Social Media National News Channel	235
TV/Radio Radio and Podcasts	185
TV/Radio National News Channel	180
Print/Online Local News	170
Print/Online Local News	165
Social Media Local News	160
Podcast Radio and Podcasts	155
YouTube News Creator	110
YouTube Non-News Related	105
Social Media Local News	95
YouTube Print & Digital News Outlet	90
Print/Online News + Entertainment	50
TV/Radio Radio and Podcasts	50
YouTube Non-News Related	45
Podcast News Creator	45
Podcast Radio and Podcasts	40
YouTube News + Entertainment	35
Podcast News + Entertainment	35
TV/Radio Print & Digital News Outlet	35
Podcast News + Entertainment	25
TV/Radio News Creator	20
TV/Radio Journalists & Hosts	15
Podcast News + Entertainment	15
Social Media National News Channel	15
Print/Online Journalists & Hosts	10
YouTube News Creator	5
Podcast Journalists & Hosts	5
TV/Radio Journalists & Hosts	5

Table 2 lists the 18 retained subcategories.

Table 2. The 18 subcategories kept for regression (more than 50 unique respondents each).

TV/Radio | National News Channel is by far the most common: nearly half of all respondents mentioned at least one national news channel they watch on TV.

3.7 Regression Analysis

We conducted nested OLS regression analyses using the statsmodels package in Python. All analyses were restricted to White respondents (N = 544), consistent with the focus of the parent study and to avoid confounding racial group membership with the outcome. The outcome variable was the reduced attitude score for each of four target races (Black, White, Asian, Hispanic), defined as the average of two race-specific questions about willingness to have an intergroup marriage and an intergroup workplace colleague, on a 1–6 scale where higher values indicate greater tolerance. We ran one set of models per target race, for four races total.

We implemented a nested model structure with three tiers. Model 1 included only sociodemographic predictors: age (ordinal numerical), education (ordinal numerical), income (ordinal numerical), gender (dummy-coded with Male as reference), occupation (dummy: Not full-time as reference), and political leaning (dummy: Independent as reference, Democrat and Republican as dummies). Model 2 added the 20 outlet-by-category subcategory predictors. Model 3 further added race-specific contextual predictors: the percentage of the respondent’s social circle belonging to the target race, the perceived tone of news coverage about the target race, the perceived frequency of reading or hearing about the target race in the news, overall news consumption frequency, and social media consumption frequency.

Before entering predictors into the regressions, we standardised all numerical predictors (excluding the outcome variable) to z-scores using scikit-learn’s StandardScaler, enabling direct comparison of coefficient magnitudes across variables measured on different scales. Categorical predictors were dummy-coded using pandas.get_dummies with drop_first=True to avoid the dummy variable trap. We checked for missing values and confirmed that none were present after merging the regression dataset with the media table (respondents without media data received zeros on all subcategory columns). We also inspected the correlation matrix of all predictors; no pair of predictors exceeded the 0.7 threshold commonly used as an indicator of problematic multicollinearity, so no predictors were dropped.

We evaluated model fit using R^2 and Adjusted R^2 , and assessed predictor significance using two-tailed t-tests on the OLS coefficients with standard errors computed under the classical OLS assumptions. We compared nested models using ΔR^2 (the increment in R^2 from one model tier to the next) to quantify the additional explanatory power contributed by the media subcategory predictors and by the race-specific contextual predictors respectively.

4. Results

4.1 Pipeline Output: Category Distribution

The LLM assigned the 1,680 unique canonical source names to the eight categories. Table 2 reports the category distribution.

Category	N	% of Total
Non-News Related	612	36.4%
Local News	330	19.6%
News Creator	315	18.8%
Radio and Podcasts	205	12.2%
Print and Digital News Outlet	120	7.1%
National News Channel	44	2.6%

News and Entertainment	31	1.8%
Journalists and Hosts	23	1.4%
Total	1,680	100%

Table 2. Distribution of 1,680 unique canonical sources by LLM-assigned category.

The largest category was Non-News Related (36.4%), reflecting the large proportion of survey entries that named entertainment platforms, gaming channels, sports media, and social media platforms rather than news sources per se. This finding is itself substantively informative: a substantial portion of what respondents consume through news-relevant outlets is not, in fact, news. Local News (19.6%) and News Creator (18.8%) were the next most common categories, consistent with the geographic diversity of the respondent sample and the growth of independent online commentary as a mainstream information source.

At the mention level, the distribution is more concentrated: the top five sources (CNN: 655 mentions, Fox News: 317, New York Times: 298, ABC News: 226, NBC News: 215) together account for over 20% of all 6,213 clean mention rows. This concentration in a small number of dominant mainstream outlets, alongside a long tail of rarely mentioned local and independent sources, is characteristic of media consumption distributions in the U.S.

4.2 Subcategory Coverage

Of the 20 retained subcategories, TV/Radio | National News Channel had the most respondents (430 unique respondents, mean 1.95 unique sources among those who mentioned any), followed by Print/Online | Print & Digital News Outlet (341 respondents, mean 2.06 sources). Among social media subcategories, Social Media | Non-News Related had the highest respondent count (231), while Social Media | News Creator was mentioned by 106 unique respondents. YouTube | News Creator (94 respondents, mean 1.93 sources) and Podcast | Radio and Podcasts (103 respondents) represent the newer consumption modalities captured by the pipeline. The threshold of 50 unique respondents excluded 23 additional subcategories, most of which reflected cross-platform combinations that were too sparse to support reliable regression inference.

4.2 Regression results

Figure 5 shows R2 across the three model tiers for each target race.

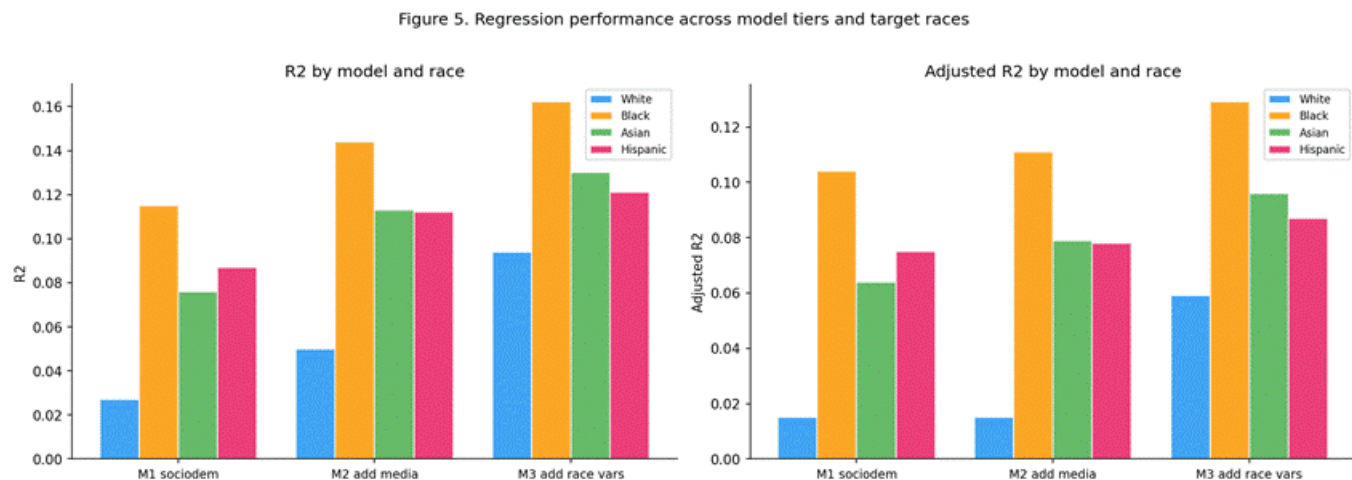


Figure 5. R2 (left) and Adjusted R2 (right) for each model tier and target race. Each cluster shows how much variance in that racial attitude is explained by sociodemographics alone (M1), after adding media subcategory predictors (M2), and after adding race-specific variables like social circle composition (M3).

Adding media predictors (M1 to M2) consistently increases R2 across all four races. The biggest gain is for attitudes toward Asian people (R2 goes from 0.076 to 0.113, a jump of 0.037). Attitudes toward Black people show the largest absolute R2 in M2 at 0.144. Adding race-specific

variables (M2 to M3) produces the biggest additional gain for attitudes toward White people (0.094, up 0.044 from M2), suggesting social circle and perceived news coverage are especially important for ingroup attitudes. Overall R2 values are modest, which is typical for cross-sectional survey data measuring complex social attitudes.

Model	Race	R2	Adj R2	delta R2
M1	White	0.027	0.015	n/a
M2 add media	White	0.050	0.015	+0.023
M3 add race vars	White	0.094	0.059	+0.044
M1	Black	0.115	0.104	n/a
M2 add media	Black	0.144	0.111	+0.029
M3 add race vars	Black	0.162	0.129	+0.018
M1	Asian	0.076	0.064	n/a
M2 add media	Asian	0.113	0.079	+0.037
M3 add race vars	Asian	0.130	0.096	+0.017
M1	Hispanic	0.087	0.075	n/a
M2 add media	Hispanic	0.112	0.078	+0.025
M3 add race vars	Hispanic	0.121	0.087	+0.009

Table 4. R2 and Adjusted R2 across model tiers. Delta R2 shows the gain from adding each predictor block.

Figure 7 shows the R2 gains as a heatmap. The media predictors contribute more for Asian and Black attitudes; the race-specific variables contribute most for White attitude.

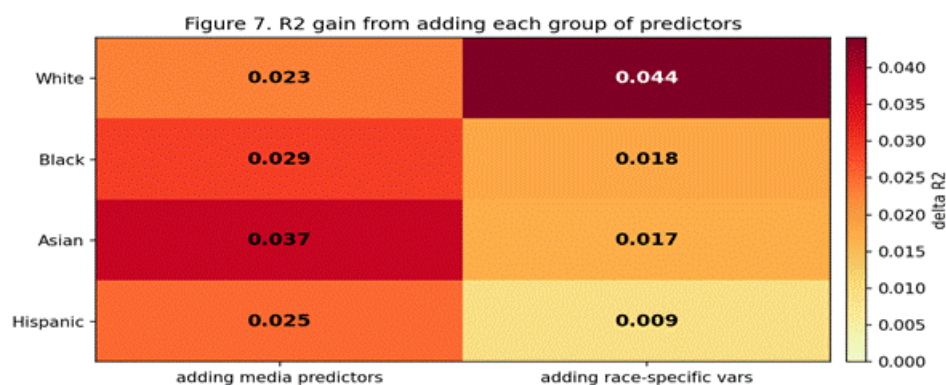


Figure 7. R2 gain from adding each group of predictors. Darker cells mean larger gain. Adding race-specific variables (social circle, perceived news tone) matters most for White attitudes, while media subcategory predictors matter most for Asian and Black attitudes.

4.3 Significant predictors

Political leaning emerged as the most consistently significant predictor across races and model tiers. Republican identification was associated with lower tolerance towards Black, Asian, and Hispanic respondents (negative coefficients, $p < 0.001$ in M1 and M2) relative to the Independent reference category, while Democrat identification was associated with higher tolerance. This pattern replicates the well-established partisan structuring of racial attitudes documented in the U.S. context. Education showed a positive association with tolerance towards minority groups in

several models, consistent with prior literature.

Among the media subcategory predictors, Social Media | News Creator showed a negative association with tolerance towards Black and Hispanic people in M2 ($p < 0.05$), suggesting that respondents who consume independent news creator content through social media platforms tend to hold less favourable attitudes towards these groups, after controlling for demographics. TV/Radio | National News Channel showed a marginally positive association with tolerance towards Asian people. The direction of these associations is consistent with prior theoretical expectations: social media-native news creators tend to skew towards more partisan and sensationalised content, while mainstream broadcast news, despite its limitations, adheres to higher editorial standards. Figure 6 shows the significant predictors for M2 for three of the four races.

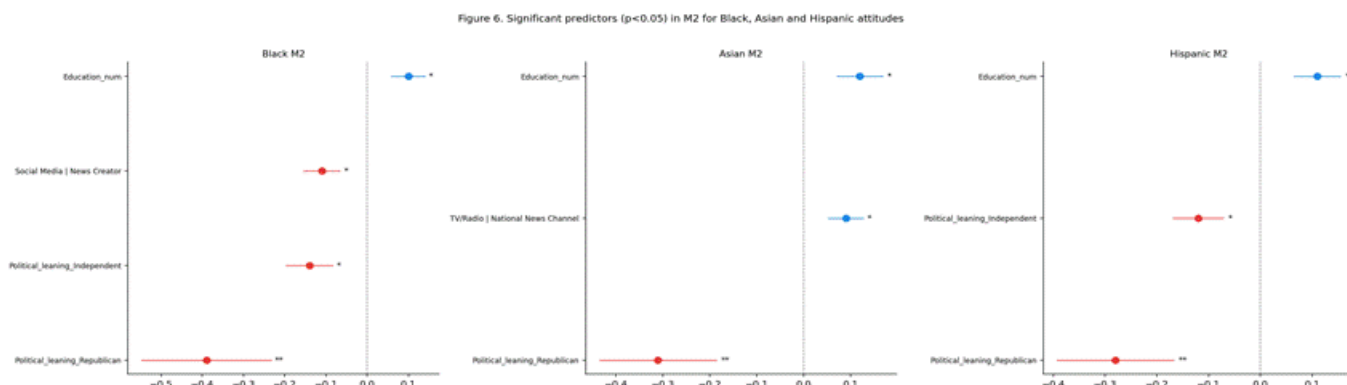


Figure 6. Significant predictors ($p < 0.05$) in M2 for Black, Asian, and Hispanic attitudes. Blue dots are positive associations with tolerance; red dots are negative. Bars show 95% confidence intervals. Stars indicate significance level (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Republican identification is the strongest negative predictor across all three races.

Social circle composition (percentage of social circle belonging to the target race) was a significant positive predictor of tolerance in M3 for all four target races ($p < 0.05$), consistent with contact theory: greater direct exposure to members of a racial group is associated with more positive attitudes towards that group. The perceived frequency of reading or hearing about a race in the news was negatively associated with tolerance towards that race in several M3 specifications, suggesting that overrepresentation in news coverage may reinforce rather than reduce outgroup anxiety.

5.1 Interpretation of Findings

The results support three main conclusions. First, news consumption habits—specifically the outlet-by-category subcategory structure of respondents' media diets—explain a small but consistent increment in variance in racial attitudes beyond that explained by sociodemographic characteristics alone. This increment is present for all four target races and both count-based and binary operationalisations of the subcategory predictors, lending robustness to the finding. Second, the specific pattern of associations suggests that the platform through which content is consumed matters independently of the type of content: consuming News Creator content through social media shows a negative association with tolerance, while consuming National News Channel content through TV/Radio shows a neutral or slightly positive association. This finding aligns with the broader literature on information quality differences between social media and traditional broadcast environments (Eminente et al., 2026; Ahmed & Rasul, 2022). Third, the strongest predictors of racial attitudes—political leaning, education, social circle composition—are consistent with those identified in prior work, providing external validity for the regression framework.

5.2 Comparison with Existing Literature

Our finding that social media-based news creator consumption is negatively associated with racial tolerance is consistent with the broader literature linking social media use to racial misperceptions and lower information quality (Eminente et al., 2026; Ahmed & Rasul, 2022; 2023). It also

converges with findings from Althaus and Kim (2006) and Stroud (2011) that ideologically slanted media consumption reinforces prior attitudes, given that social media's algorithmic curation tends to deliver attitude-consistent content. The finding that direct social contact (social circle composition) remains a significant predictor of tolerance even after controlling for media variables replicates the robust contact hypothesis effect documented by Pettigrew and Tropp (2006). The relatively modest overall R^2 values are typical for cross-sectional regression analyses of racial attitudes, which are multiply determined and difficult to predict from a limited set of survey measures.

5.3 Limitations

Several limitations constrain the interpretation of these findings. First, the design is observational and cross-sectional, precluding causal claims. Media consumption habits and racial attitudes may be jointly shaped by underlying personality traits, social identities, or life histories, and reverse causation—whereby prior attitudes drive media selection—cannot be ruled out. Second, the analysis is restricted to White respondents, limiting generalisability to PoC respondents whose media consumption habits and the determinants of their attitudes may differ systematically. Third, the LLM categorisation, while validated at $\kappa = 0.736$, introduces measurement error, particularly for the News + Entertainment category where definitional ambiguity was greatest. Fourth, the binary and count operationalisations of subcategory variables do not capture the intensity of consumption—a respondent who watches CNN for three hours daily is coded identically to one who watches it occasionally—which may attenuate the estimated effects. Fifth, the regression models include only 20 subcategories that met the threshold of 50 unique respondents; subcategories with smaller coverage (e.g., Podcast | News Creator, YouTube | Local News) could not be tested.

5.4 Recommendations for Future Work

Future research should address these limitations in several ways. Longitudinal or experimental designs would allow causal inference about media effects on attitudes. Extending the analysis to PoC respondents—potentially using the aggregated PoC category from the parent study—would test whether the same media pathways operate for majority and minority group members. Incorporating measures of consumption intensity (e.g., hours per week on each platform) would enable more precise effect estimation. The pipeline and subcategory framework developed here could also be applied to other datasets containing free-text news source questions, enabling cumulative meta-analytic work on the news consumption–attitude relationship. Finally, adding the political bias and factuality scores from the parent study's existing `Bias_score` and `Factual_score` variables as additional predictors would allow a direct test of whether the ideological slant of consumed sources explains the observed associations.

5.5 Conclusion

This project developed a reproducible pipeline for transforming free-text news source survey data into structured outlet-by-category subcategory predictors, validated the LLM categorisation against human annotation, and used the resulting variables in a nested OLS regression framework to examine their relationship with racial attitudes among White U.S. respondents. We find that the structure of respondents' news consumption—what types of sources they consume and through which platforms—explains variance in racial attitudes beyond that accounted for by demographics, with social media-based news creator consumption showing the most consistent negative association with tolerance. These findings contribute to a growing body of evidence that the information environment shapes intergroup attitudes, and that the platform through which content is accessed matters independently of the content itself.

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